

DEQ SITE IDENTIFICATION FORM

Complete the following information and provide the attachments listed below for each new site or existing site that has not had a Site Identification Form with attachments submitted to the DEQ since January 1, 1998. Please amend any previous Site Identification Forms when it becomes known that any of the information requested below has changed (i.e. new land owner).

FACILITY NAME City of Cadillac WWTP		NPDES or COC PERMIT NUMBER MI0020257	
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SITE IDENTIFICATION FORM

A. Site ID	Site ID # 21N08W30-AB01	Street or Route # East of Seeley RD, South of 48 RD		
B. County/Township:	County Missaukee		Township Richland	
C. State Planar Coordinates:	NW 1/4	SW 1/4	Section 30	Town T21N
			Range R08W	
D. Lat/Long (nearest second):	Latitude 44° 11' 22.43"N		Longitude 85° 19' 58.23"W	
			Method Used (i.e. GPS, topo) Google Earth	

E. Site Ownership	Name of Owner Alan Benson		Phone 231-920-6783	
	Street or P.O. Box 11799 East 50 Mile Rd			
	City or Town Cadillac		State MI	Zip 49601
	Is there a written agreement from the owner to land apply? ☒ Yes ☐ No, written agreement is required		Previous landowners name Hartley Benson	
			Property purchase date N/A	

	Name of Lessee (if different) Mike Bosscher		Phone 231-499-4256	
	Street or P.O. Box 11320 W Stoney Corners Rd.			
	City or Town McBain		State MI	Zip 49657

F. Site Characteristics	Site Type ☒ Agricultural ☐ * Reclamation Site ☐ Forest ☐ ** Lawn/Home Garden *Separate approval is required **Requires EQ Biosolids		Soils Information Date of last soils analysis 11-15-2019 *Phosphorous 65 units PPM Potassium 97 units PPM pH 7.0 Soil type(s) sandy *300 lb/ac Bray P1 maximum							
	Expected crops/vegetation to be grown <table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 40%;">Crop/Vegetation Corn</td> <td>Nitrogen Requirement 135 lbs. N/acre</td> </tr> <tr> <td>_____</td> <td>_____ lbs. N/acre</td> </tr> <tr> <td>_____</td> <td>_____ lbs. N/acre</td> </tr> </table>		Crop/Vegetation Corn	Nitrogen Requirement 135 lbs. N/acre	_____	_____ lbs. N/acre	_____	_____ lbs. N/acre		
Crop/Vegetation Corn	Nitrogen Requirement 135 lbs. N/acre									
_____	_____ lbs. N/acre									
_____	_____ lbs. N/acre									
	Total acreage of site 18.2 Acreage used for crops 18.2		Percent slope Highest percent slope of the site? ☐ 0-6% ☐ 6-12% ☒ *Higher (Requires an approved site reclamation plan)							
	Public Exposure Is this land with a high potential for public exposure? (See Definitions section on Page 19) ☐ Yes ☒ No		If agricultural, is the site tilled? ☐ Yes ☐ No ☒ NA							

On an attached sheet provide a copy of the following:

- A Farm Services Agency map (formerly ASCS) of the proposed application site with property boundaries, surface waters, and discharge location of field tiles when present.
- A copy of the soil test analysis that is no older than 2 years old at the time of land application.
- A copy of the plat map (or other map showing property dimensions) with the proposed land application site highlighted.
- A soil survey map with the land application site identified.



CITY OF CADILLAC WASTEWATER TREATMENT PLANT

BIOSOLIDS RECYCLING PROGRAM PARTICIPATION

I hereby agree to participate in the City of Cadillac WWTP biosolids recycling program. I understand that biosolids will be applied to my land in accordance with federal EPA regulations and guidelines of the Michigan Department of Environmental Quality.

I understand that biosolids have been analyzed to be sure they can be applied to agricultural land in accordance with federal regulations and MDEQ guidelines. I understand that biosolids contain some heavy metals and that they must be below the EPA allowable limits for land application. I understand that the nutrients and/or the metal concentration may limit the frequency of biosolids application to my land.

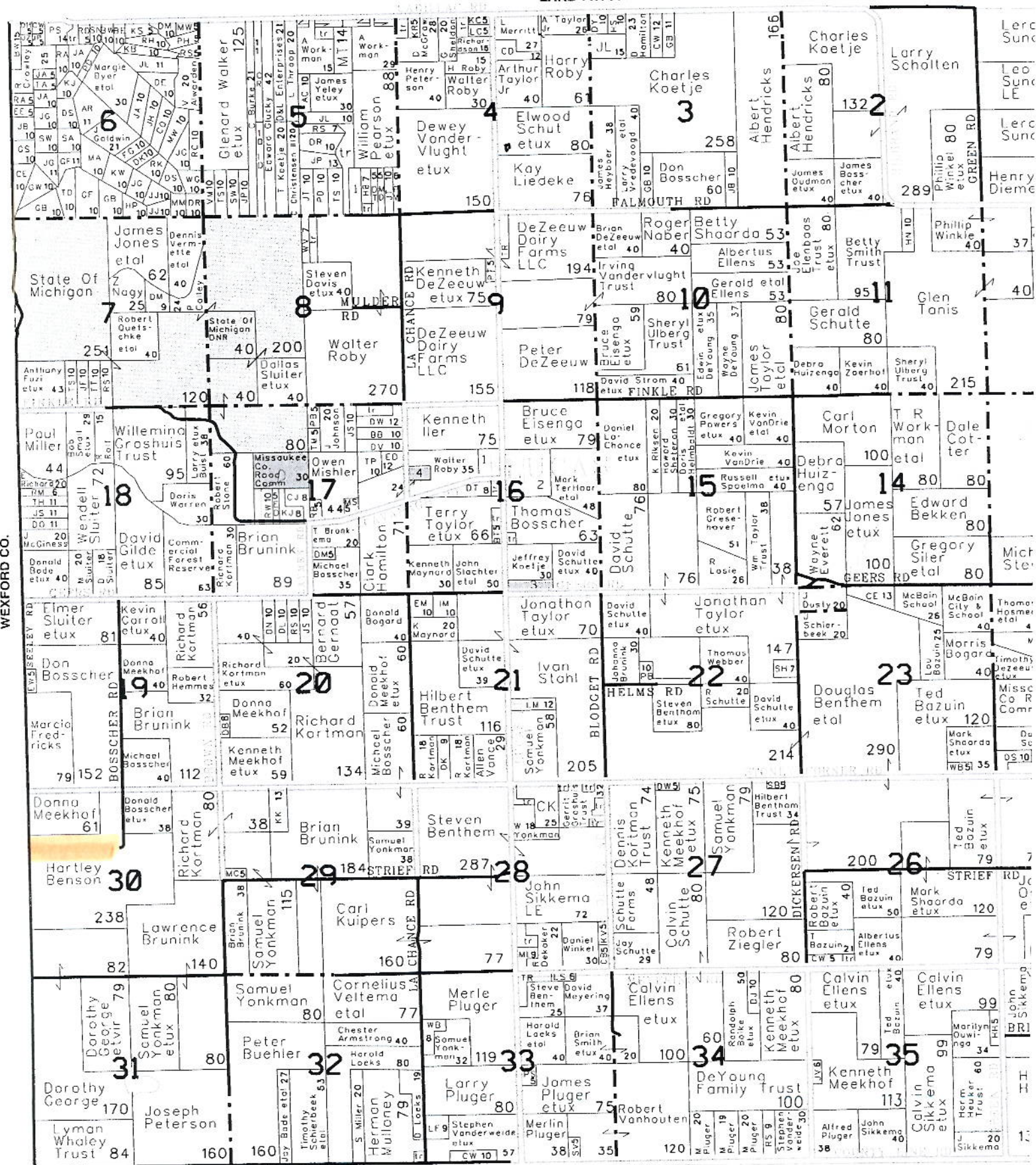
My land is used for the production of agricultural crops. Information concerning crop and site restrictions has been explained to me. I agree to allow representatives of the MDEQ, Cadillac Waste Water Treatment Plant and the City's Injection contractor to have access to my land to obtain soil samples and for application of biosolids.

If I decide to cancel this agreement, I will give 60 days advance notice to the City of Cadillac WWTP.

Landowner: Alan Benson Signed: Alan Benson
Address: 11799 Eastson Rd Date: 10-29-19
Township: Richland Section: 30 Acres: 18.2
City WWTP Field Number: 21N08W30-AB01
If Rented, Farmer: Mike Bosscher
Address: 11320 W Stoney Corners Rd.
McBain MI 49657

(Landowners)

LAKE TWP, PAGE 30



Soil Map—Missaukee County, Michigan
ZINOSW30--AB01



Soil Map may not be valid at this scale.

Map Scale: 1:2,250 ft printed on A landscape (11" x 8.5") sheet



MAP LEGEND

 Area of Interest (AOI)	 Spoil Area
 Soils	 Stony Spot
 Soil Map Unit Polygons	 Very Stony Spot
 Soil Map Unit Lines	 Wet Spot
 Soil Map Unit Points	 Other
Special Point Features	 Special Line Features
 Blowout	Water Features
 Borrow Pit	 Streams and Canals
 Clay Spot	Transportation
 Closed Depression	 Rails
 Gravel Pit	 Interstate Highways
 Gravelly Spot	 US Routes
 Landfill	 Major Roads
 Lava Flow	 Local Roads
 Marsh or swamp	Background
 Mine or Quarry	 Aerial Photography
 Miscellaneous Water	
 Perennial Water	
 Rock Outcrop	
 Saline Spot	
 Sandy Spot	
 Severely Eroded Spot	
 Sinkhole	
 Slide or Slip	
 Sodic Spot	

MAP INFORMATION

The soil surveys that comprise your AOI were mapped at 1:15,800.

Warning: Soil Map may not be valid at this scale.

Enlargement of maps beyond the scale of mapping can cause misunderstanding of the detail of mapping and accuracy of soil line placement. The maps do not show the small areas of contrasting soils that could have been shown at a more detailed scale.

Please rely on the bar scale on each map sheet for map measurements.

Source of Map: Natural Resources Conservation Service
Web Soil Survey URL:

Coordinate System: Web Mercator (EPSG:3857)

Maps from the Web Soil Survey are based on the Web Mercator projection, which preserves direction and shape but distorts distance and area. A projection that preserves area, such as the Albers equal-area conic projection, should be used if more accurate calculations of distance or area are required.

This product is generated from the USDA-NRCS certified data as of the version date(s) listed below.

Soil Survey Area: Missaukee County, Michigan
Survey Area Data: Version 18, Sep 16, 2019

Soil map units are labeled (as space allows) for map scales 1:50,000 or larger.

Date(s) aerial images were photographed: May 9, 2014—Sep 19, 2016

The orthophoto or other base map on which the soil lines were compiled and digitized probably differs from the background imagery displayed on these maps. As a result, some minor shifting of map unit boundaries may be evident.

Map Unit Legend

Map Unit Symbol	Map Unit Name	Acres in AOI	Percent of AOI
11B	Montcalm-Graycalm complex, 0 to 6 percent slopes	1.2	6.4%
11C	Montcalm-Graycalm complex, 6 to 12 percent slopes	14.2	77.8%
11E	Montcalm-Graycalm complex, 12 to 30 percent slopes	2.9	15.8%
Totals for Area of Interest		18.2	100.0%

**MICHIGAN STATE UNIVERSITY
SOIL AND PLANT NUTRIENT LABORATORY
EAST LANSING, MICHIGAN 48824-1325
(517) 355-0218**

SOIL TEST REPORT FOR:				CONSULTANT			
CITY OF CADILLAC 200 N. LAKE ST. CADILLAC MI 49601				WEXFORD COUNTY MSUE 401 N. LAKE STREET83 CADILLAC MI 49601 231-779-9480			
DATE	LAB #	COUNTY	Previous Crop	ACRES	FIELD ID	SOIL	TEXTURE
11/15/2019	240355	Wexford	Corn silage	18	21n08w30-ab01	Mineral	

SOIL NUTRIENT LEVELS			Below Optimum	Optimum	Above Optimum
¹ Soil pH	7.0	Lime Index			
² Phosphorus (P)	65	ppm	■■■■■■■■■■■	■■■■■■■■■■■	■■■■■■■■■■■
³ Potassium (K)	97	ppm	■■■■■■■■■■■	■■■■■	
³ Magnesium (Mg)	71	ppm	■■■■■■■■■■■	■■■■■■■■■	

ADDITIONAL RESULTS:					Optional Tests:						
Calcium (Ca) (ppm)	CEC (meq/100 g)	% of Exchangeable Bases			Micronutrients (ppm)					Organic Matter %	Nitrate-N ppm
		K	Mg	Ca	B	Cu	Mn	Zn	Fe		
936	5.5	4.5	10.7	84.8							

RECOMMENDATIONS:

Limestone: NONE

Tillage Depth: 9 inches

Target pH = 6.5

% Stand:

Plant Nutrients:					Micronutrient: (Optional)				
Year	Crop	Expected Yield	Nitrogen (lb N/A)	Phosphate (lb P ₂ O ₅ /A)	Potassium (lb K ₂ O/A)	Boron (lb B/A)	Manganese (lb Mn/A)	Zinc (lb Zn/A)	Copper (lb Cu/A)
1	Corn silage	18 ton	135	0	145	0.0			

Use a starter fertilizer high in nitrogen (30-40 lb N/A) when planting into heavy residue.

For questions about interpreting your soil results go to the following website: <http://msue.anr.msu.edu/experts> and contact the MSUE field crops agent in your area.

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dlangworthy@cadillac-mi.net

Test Methods: 1- 1:1 soil:water pH, 2- Bray P1 Extractant, 3- 1N Ammonium Acetate Extractant